Problem A. Breakingsky and their common

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

BreakingSky is a team consisting of three members, whose names are *Pegasus*,*Hitass_SY* and *Nekopass*.



Hitass_SY is always interested in the team's signature. The signature of a team is defined as **the longest common subsequence** of all the members' names of the team. For example, if a team consists of three members named *Psusage*, *Hssati_SY* and *Nssapoke*, then the team's signature is *ssa* because it is the longest common subsequence of *Psusage*, *Hssati_SY* and *Nssapoke*.

Now Nekopass receives a 5-line message from Hitass_SY. To figure out what information Hitass_SY intended to deliver, Nekopass first needs to delete all the spaces in the message. Then Nekopass will count how many times the signature of BreakingSky appears in the message as a substring **ignoring case in each line** and then **add them up**. But Nekopass is busy playing Genpact Imshin, so he asks you to find

out the answer.

To prevent you from making mistakes, Nekopass reminds you that the names of BreakingSky's three members are *Pegasus*,*Hitass_SY* and *Nekopass*, and a team's signature is **the longest common subsequence** of the team's members' names.

A ***substring*** of a given sequence is a consecutive sequence of characters occurrences at least once in the given string.

A ***subsequence*** of a given sequence is a sequence that can be derived from the given sequence by deleting some or no elements without changing the order of the remaining elements.

Input

5 lines of strings. It is guaranteed that the number of characters input in each line would not exceed 10^5.

Output

One integer, the answer to the input strings.

Example

standard input	standard output
Breakingsky has three members. They are pegasus , hitass_SY and nekopass. They are members in ACMM. ACMM,YES! I love SYSU , just as SYSU loves me.	3

Note

To read a whole line in C++, you may need cin.getline()

Problem B. Breakingsky and graph

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

In Graph Theory we have learned how to calculate the single-source shortest path in a graph with n vertices and m edges, but Pegasus thinks that all the shortest path algorithms running on limited graph are useless, and "completely inextensible".

What you said is right, but now I'm creating an Open World Game. The game happens in a fantasy world called "Sun Yeah Sand", where selected people are granted "Sun's Eye ", and will guide the power of Sun. In the end of the journey, you will finally discover the ultimate truth of "Sun Yeah Sand".....

"Sun Yeah Sand" is an infinite open world with infinite city. We can label the city in this world 0, 1, 2, 3, ... , and for each city p, there are three directed roads starting from p:

- A directed road from p to $p + 1$ with length 1.
- A directed road from p to $p * 3$ with length 1.
- A directed road from p to $p * 4$ with length 1.

Now you are a traveler and decide to take T journeys. In the i-th journey you will start at city 0 and end at city x_i, and you need to calculate the minimum distance you have to travel.

Input

The first line contains one integer T ($T \leq 10$).

Then follows T lines, the i-th line contains one integer x_i ($0 \leq x_i \leq 10^9$).

Output

T lines, the i-th line representing the answer of the i-th journey.

Example

standard input	standard output
7 0 1 2 3 4 5 6	0 1 2 2 2 3 3

Problem C. Breakingsky and space

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

In the game Honor of Kings, there is a hero, Genghis Khan, who is almost forgotten by everyone. His second skill Beast Trap can significantly increase the damage dealt by Genghis Khan if used properly. As you can see, the trap's trigger radius is far smaller than its damage radius, as shown in the picture, which means that if someone triggers the trap, one may be attacked by the trap even if he is not in the trigger radius of the trap.

Taking inspiration from life, Hitass_ SY invents a trap sphere in three-dimensional space. The trap sphere has a damage radius R and an infinite trigger radius. Now Hitass_ SY places n trap sphere in the three-dimensional space, and the i-th trap sphere's sphere center is at (X_i, Y_i, Z_i) and can deal B_i damage to each one whose distance from the sphere center is no bigger than the damage radius once the trap sphere explodes. However, due to the difficulty of researching, all the trap spheres share the same damage radius R.



Now there are m spaceships flying along some straight lines. When an airship visits a point (x, y, z) and x, y, and z are all integers, the spaceship will leave one passenger at that point. Given the spaceship's position vector (x_i, y_i, z_i) and direction vector (a_i, b_i, c_i), the airship will visit point

$$(x_i + ka_i, y_i + kb_i, z_i + kc_i), k \in \mathbb{R}$$

Once all the passengers have been left, all the traps will explode and deal damage to those passengers who are in the damage radius. There are infinite passengers left in the three-dimensional space so Hitass_SY

does not want to collect data about them. So now you have to help Hitass_SY calculate the total damage taken by all the passengers left by spaceships.

Input

The first line contains three integers, n, m, R.($0 < n, m \leq 1000$; $0 < R \leq 10000$)

Then follows one line with n integers. The i-th integer represents B_i.

Then follows n lines. The i-th line contains three integers X_i, Y_i, Z_i.

Then follows m lines. The i-th line contains six integers x_i, y_i, z_i, a_i, b_i, c_i. ($a_i^2 + b_i^2 + c_i^2 > 0$)

The absolute value of X_i, Y_i, Z_i, x_i, y_i, z_i, a_i, b_i, c_i would not exceed 10000.

Output

One integer, the total damage taken by all the passengers left by spaceships.

Example

standard input	standard output
1 1 5000 1 0 0 0 0 0 0 1 0 0	10001

Problem D. Breakingsky and star rail

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

Nekopass, together with his friends, takes Astral Express across the Milky Way, along the journey of some Aeon, towarding the end of the journey.

The galaxy has n planets, and Akivili has left m rails among the n planets. Each rail starts at some planet u and ends at some planet v and has its own length w, which means that you can travel from planet u to planet v quickly by using this rail. Notice you *cannot* travel from planet v to planet u by using this rail. However, using rails is not free, because you should pay Pom-Pom, the conductor, some Stellar Jade for using a rail, and as the number of rails you have used grows, the cost also grows. Specifically, if you use k rails in the order of p_1, p_2, ... p_k, then you should pay $\sum_{i=1}^{k} w_{p_i}^i$ Stellar Jade to Pom-Pom.

Nekopass is planning to spend as much as possible Stellar Jade for gacha, so he should save money when traveling. Now Nekopass wants to know with K Stellar Jade, whether he can travel from planet S to planet T. If he can't afford the travel, output 0. Otherwise, output the maximum Stellar Jade he can reserve after traveling from S to T.

Input

The first line contains five positive integers, n, m, K, S, T. ($S, T \leq n \leq 50000$, $m \leq 200000$,$K < 2^{30}$)

Then follows m lines. The i-th line contain three integers, u_i, v_i and w_i, ($1 \leq u_i, v_i \leq n$, $1 < w_i \leq 2^{30}$)indicating that the i-th rail connect planet u_i, v_i with length w_i. It is guaranteed that there are no self-loops or multiple edges.

Output

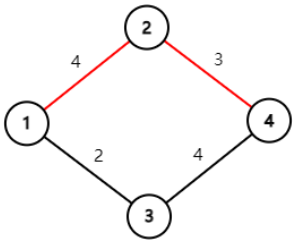
One integer, denoting the answer.

Example

standard input	standard output
4 4 64 1 4 1 2 4 2 4 3 1 3 2 3 4 4	51

Note

In the example,



The lowest cost is $4^1 + 3^2 = 13$ by visiting planets 1, 2, 4 in order.

So Nekopass can reserve at most $64 - 13 = 51$ Stellar Jade.

Problem E. Breakingsky and ping_fan

Input file:	standard input
Output file:	standard output
Time limit:	1 second
Memory limit:	256 megabytes

Ping_fan is a good friend of BreakingSky. He loves playing a music game called "Project phigarcea: chunimaiDX". The level of a player can be measured by a positive integer known as *potential* (*ptt* for short).

The difficulty of each song in the game is decided by three factors, the song's speed x, the density of note y ($x, y \in \mathbb{C}$), and the rank of the song n. Similarly, the song's level is defined as follows

$$level = x^n + y^n$$

Here we define $0^0 = 1$.

To enhance his strength in playing music games, Ping_fan will choose a song whose level is the same as his *ptt* to play. However, because of the strange ideas of the developers, the game shows $A = x+y$, $B = xy$ and n (it is guaranteed that A, B are non-negative integers), instead of x, y and n, because the song's speed x and density of notes y may be **complex numbers**. Now given a song's A, B and n, Ping_fan wants you to tell with how much *ptt* will he choose this song.

Notice: Considering it may be hard to know what x and y exactly are and calculate x^n and y^n, you'd better find a way to calculate $x^n + y^n$ without knowing x and y.

Input

The first line contains one positive integer T ($T \leq 1000$).

Then follows T lines, each line contains three non-negative integers A, B ($0 \leq A, B < 10^9 + 7$) and n ($\sum n \leq 1000000$), denoting a song's information shown in the game.

Output

Output T lines. Each line contains one integer denoting the answer module $10^9 + 7$.

Example

standard input	standard output
3 2 2 2 1 1 1 0 0 0	0 1 2

Note

In the example,

- For the first song, $x = 1 + i$ and $y = 1 - i$, so $x^2 + y^2 = 0$.
- For the second song, $x = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $x = \frac{1}{2} - \frac{\sqrt{3}}{2}i$, so $x^1 + y^1 = 1$.
- For the first song, $x = 0$ and $y = 0$, so $x^0 + y^0 = 2$.

Problem F. Breakingsky and sequences

Input file:	standard input
Output file:	standard output
Time limit:	2 seconds
Memory limit:	256 megabytes

Hitass_SY is solving classic problems these days. One day he saw a problem: Given a non-decreasing sequence a_1, a_2, ... a_n, you are asked to find the minimum $\max\limits_{1<i\leq m} |b_i - b_{i-1}|$ where b_1, b_2, ... b_m is a subsequence of $\{a_i\}$.

"You are formidable, my lordship. I have to admit. said Pegasus, who plans to make the problem harder, "But how should you respond if I eliminate the condition 'non-decreasing' of a_i from the problem?"

Hitass_SY quickly finds that he can't solve the problem on his own. So will warm-heated you help Hitass_SY solve this problem?

Input

The first line contains two positive integers n, m ($1 < m \leq n \leq 50000$).

The second line contains n integers a_i ($|a_i| \leq 1000000000$).

Output

One integer, denoting the minimum answer you get.

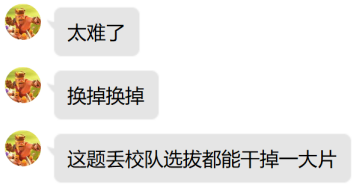
Example

standard input	standard output
3 3 5 2 7	5

Problem G. Breakingsky and warehouse

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

This problem used to be another problem but under someone's demand, it was replaced by this problem now.



Hitass_SY has been playing a boring game recently. In the game, you have K treasures and you need to place them in a rectangular treasure case with $n \times m$ grids. Each treasure occupies a rectangular area of $x_i \times y_i$ grids and has its value v_i. The game's goal is to place some treasure in the treasure case and maximize the sum of the treasures' value in the treasure case. You may **NOT** place a treasure in the treasure case after rotating it at any non-zero angle. Also, two treasures may **NOT** share a common grid.

Now Hitass_SY wants to know what is the maximum value he can get.

Input

The first line contains three integers, n, m, K. ($1 \le n \times m \le 100000$, $K \le 100000$)

Then follows K lines, each line contains three integers x_i, y_i, v_i. ($1 \le x_i, y_i \le 2$, $x_i + y_i > 2$, $0 \le v_i \le 1000$)

Output

One non-negative integer, representing the answer.

Example

standard input	standard output
2 3 3 2 2 10 1 2 2 2 1 3	13